The Mole Concept

Answer key with worked solutions

A. Understanding the mole

1. What is a mole?

> A mole is the amount of substance that contains 6.022×10^{23} particles (atoms, molecules, ions or formula units).

2. Same number of atoms?

> Yes. Both are one mole, so both contain 6.022×10^{23} atoms. The masses differ because a magnesium atom is heavier than a carbon atom.

3. What is wrong?

> Oxygen gas is made of O_2 molecules. One mole contains 6.022×10^{23} *molecules*, which is 2 atoms/molecule × 6.022×10^{23} molecules = 1.204×10^{24} oxygen *atoms*.

B. Molar mass

4. Molar masses.

> (a) CH_4: 12.01 g/mol + 4(1.008 g/mol) = **16.04 g/mol**
>
> (b) NH_3: 14.01 g/mol + 3(1.008 g/mol) = **17.03 g/mol**
>
> (c) $CaCO_3$: 40.08 g/mol + 12.01 g/mol + 3(16.00 g/mol) = **100.09 g/mol**

5. Molar masses.

> (a) $C_6H_{12}O_6$: 6(12.01 g/mol) + 12(1.008 g/mol) + 6(16.00 g/mol) = 72.06 g/mol + 12.096 g/mol + 96.00 g/mol = **180.16 g/mol**
>
> (b) $MgCl_2$: 24.31 g/mol + 2(35.45 g/mol) = **95.21 g/mol**

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C. Converting between mass, moles and particles

6. Moles in 25.0 g of $\mathrm{CH_4}$.

> $$n = \frac{m}{M} = \frac{25.0\ \mathrm{g}}{16.04\ \mathrm{g/mol}} = \mathbf{1.56\ mol}$$

7. Mass of 0.150 mol of $\mathrm{CaCO_3}$.

> $$m = n \times M = 0.150\ \mathrm{mol} \times 100.09\ \mathrm{g/mol} = \mathbf{15.0\ g}$$

8. Molecules in 0.500 mol of $\mathrm{NH_3}$.

> $$N = 0.500\ \mathrm{mol} \times 6.022 \times 10^{23}\ \mathrm{molecules/mol} = \mathbf{3.01 \times 10^{23}\ molecules}$$

9. Moles in 1.50×10^{24} molecules of water.

> $$n = \frac{1.50 \times 10^{24}\ \mathrm{molecules}}{6.022 \times 10^{23}\ \mathrm{molecules/mol}} = \mathbf{2.49\ mol}$$

10. Mass of 3.01×10^{23} sodium atoms.

> $$n = \frac{3.01 \times 10^{23}\ \mathrm{atoms}}{6.022 \times 10^{23}\ \mathrm{atoms/mol}} = 0.500\ \mathrm{mol}$$
> $$m = 0.500\ \mathrm{mol} \times 22.99\ \mathrm{g/mol} = \mathbf{11.5\ g}$$

D. Counting atoms inside compounds

11. Moles of H atoms in 2.00 mol of glucose.

> Each $\mathrm{C_6H_{12}O_6}$ contains 12 H atoms: 2.00 mol glucose × (12 mol H / 1 mol glucose) = **24.0 mol of H atoms**

12. Oxygen atoms in 10.0 g of $\mathrm{CO_2}$.

> $$M(\mathrm{CO_2}) = 12.01\ \mathrm{g/mol} + 2(16.00\ \mathrm{g/mol}) = 44.01\ \mathrm{g/mol}$$
> $$n(\mathrm{CO_2}) = \frac{10.0\ \mathrm{g}}{44.01\ \mathrm{g/mol}} = 0.2272\ \mathrm{mol}$$
> $$\mathrm{O\ atoms} = 2 \times 0.2272\ \mathrm{mol} \times 6.022 \times 10^{23}\ \mathrm{atoms/mol} = \mathbf{2.74 \times 10^{23}\ atoms}$$

13. Mass of chlorine in 1.00 mol of $\mathrm{MgCl_2}$.

> 1.00 mol of $\mathrm{MgCl_2}$ contains 2.00 mol of Cl: 2.00 mol × 35.45 g/mol = **70.90 g**

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E. Challenge

14. More molecules: 10.0 g of $\mathrm{H_2O}$ or 10.0 g of $\mathrm{CO_2}$?

> $n(\mathrm{H_2O}) = \dfrac{10.0\ \mathrm{g}}{18.02\ \mathrm{g/mol}} = 0.555\ \mathrm{mol}$
>
> $n(\mathrm{CO_2}) = \dfrac{10.0\ \mathrm{g}}{44.01\ \mathrm{g/mol}} = 0.227\ \mathrm{mol}$
>
> **Water** contains more molecules (3.34×10^{23} compared with 1.37×10^{23}). Water molecules are lighter, so the same mass contains more of them.

15. Identify the element.

> $M = \dfrac{m}{n} = \dfrac{5.00\ \mathrm{g}}{0.125\ \mathrm{mol}} = 40.0\ \mathrm{g/mol}$. This matches **calcium** (Ca = 40.08 g/mol).

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